What Leaves With Them

When someone leaves, their team lead has to find out who owns each system now, and why. The answer is spread over chat, tickets and documents that disagree and keep changing.

67%

of 1,000 IT managers at large UK and US organizations worry about losing knowledge when people leave

64%

say their organization has already lost knowledge that way

Who it is for

A team lead handing over a leaver's work, or the new owner of a system. They ask in plain words: "Who owns billing-api now?" "Who decided ADR-0014?"

WHAT IT GIVES THEM

An answer they can check

Every cited claim comes with the record it rests on and the exact line, read again at its source before the answer is given.

A warning when a record changed

If a message was edited after the agents read it, the answer says so and flags it for a person. In our test the edited message was caught in 5 of 5 live runs.

The gaps, in writing

"fernhill-sso has no current owner. ... Nobody has been recorded as taking it over." Better to learn that before the last day than after.

Survey: Sinequa press release, August 2022 (1,000 IT managers, UK and US)

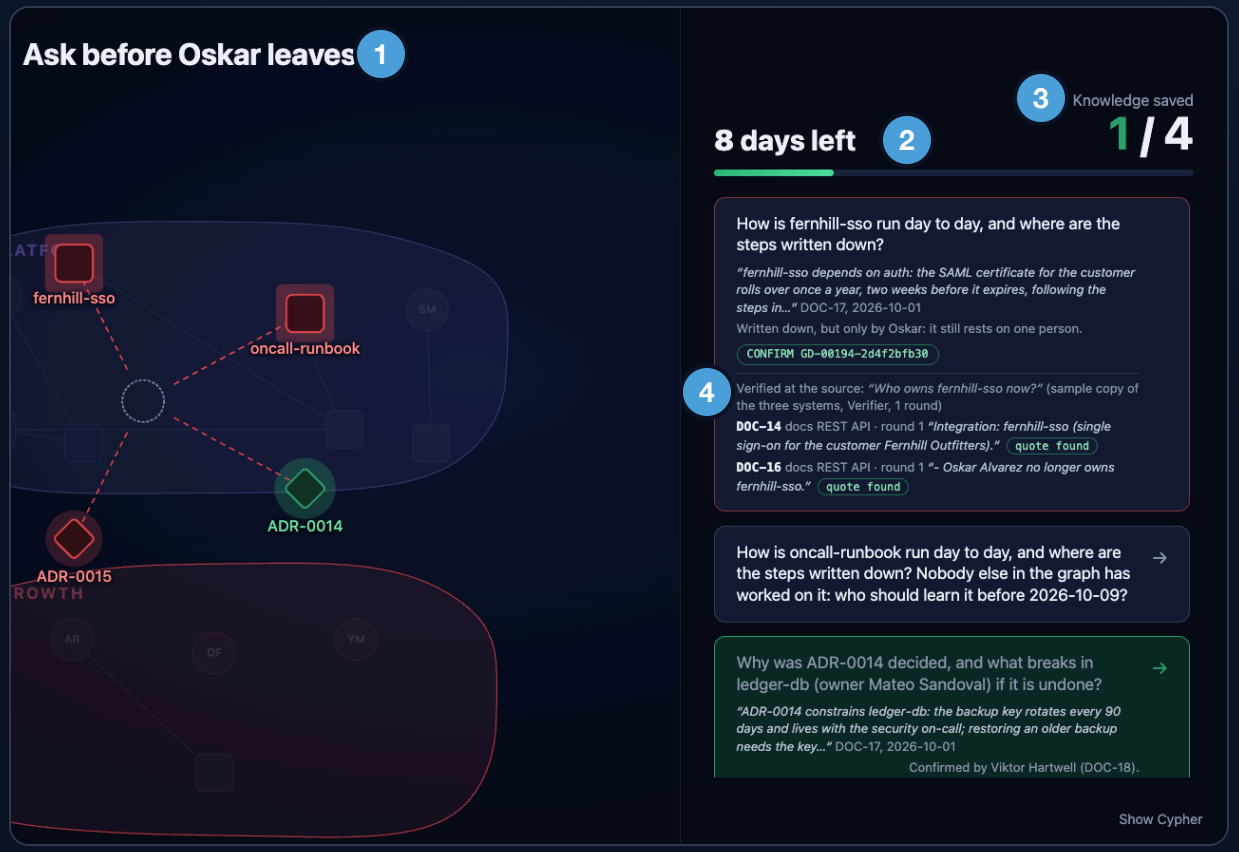
Demo data: Quillfield Billing is a made-up company. Its chat, tickets and documents come from a script (seed 42); no real people.

Two agents that check each other against the source

- **Investigator** answers the question and backs every claim with a record id and the exact line it relies on.
- **Verifier** goes back through Zetaris to the system that holds each record and checks that the line is still there. If it is not, the claim is rejected, whatever a model says.
- A rejected draft goes back with the record as it reads now. The Investigator gets at most two more tries.
- Only an answer whose every line held up is **stored**. Otherwise the answer lists what was dropped and why.

What comes back: the answer, each citation with its verdict, the dropped ones with the reason, the number of rounds, whether it was stored, and a trace of every step.

Right: a real screenshot of the panel in sample mode, cropped.



1. "Ask before Oskar leaves." Oskar Alvarez is leaving; the panel lists what rests on him alone.
2. "8 days left": a countdown from his 14-day notice, 3 days off for each question asked (a rule of the panel, not calendar days).
3. "Knowledge saved 1 / 4": 1 of the 4 items now has a second person behind it (ADR-0014, the green card).
4. The answer card, with the lines the Verifier re-read at the source (close-up on slide 7).

A loop, not a pipeline

Three systems
team chat (Postgres)
tickets (CSV file)
documents (REST API)

all three registered in Zetaris

Memory graph
built from three SELECTs through Zetaris; the write gate (FalkorDB, pre-existing) decides what is kept

H-MEM adds earlier verified answers

Investigator
21 graph tools → a draft answer with one quoted line per claim; Nemotron through Nightshift

draft ↓

Verifier
re-reads each record by id through Zetaris → is the quoted line still there? (plain code, final) → model judgement (Nemotron through Nightshift)

Rejected → the Investigator gets the notes and each record as its source holds it now → revises (at most 2 revisions, so at most 3 rounds) → Verifier again

Every line held up
stored as a finding and as an H-MEM memory

Some lines did not
the answer says what was verified and what was dropped, with the reason; nothing stored

Named in no system
one query across all three systems finds no record → the answer says so (absence_confirmed); nothing stored

> Why a loop: a model can misquote a record or answer from an old copy. The loop turns that into a revision, and a quote that fails the check is never stored.

Runs with one command; no keys in sample mode

LAYER	WHAT	WHERE IN THE REPO
Data access	Zetaris 2.4.3.1, local, one SQL dialect over Postgres (JDBC), a CSV file and a REST API	`zetaris_spike/, open_agent/sources.py`
Agents and loop	Python 3.12, plain code, OpenAI-compatible client	`open_agent/`
Models	NVIDIA Nemotron 3.5 Lightning 30B-A3B (Investigator), Nemotron 3 Nano 30B-A3B (Verifier)	`docs/METERLESS.md`
Model provider	Meterless Nightshift 1.2.1, headless, loopback only, signed receipts	`open_agent/nightshift.py`
Agent memory	Meterless H-MEM reference engine, pinned, Node sidecar	`vendor/hmem/, open_agent/hmem_bridge.py`
Graph, gate, tools	FalkorDB 6.0.2 (pre-existing core)	`honest_memory/`
Run path	`docker compose up --build` · `POST /run` · sample mode on by default	`serve.py, docker-compose.yml`

Sample mode (default, for judges)

The same SQL and the same Verifier over a local copy of the three systems; no model, no keys, no network. Fresh clone with an empty .env: 5 input examples and 1 new question through POST /run, all 6 completed (3 verified, 1 absence confirmed, 2 no_claim: the sample router has no rule for that wording, a model answers it live).

Live mode

Local Zetaris (with its own admin sign-in) and Nightshift on the same machine. Measured on one Mac Studio (Apple M5 Max, 128 GB): no separate graphics card, no cloud model API.

Fresh-clone record: SUBMISSION/FRESH-CLONE-TEST.md (commit d3f4550)

Run steps: README sections 5 to 7

Zetaris for every source read, Nightshift for every model call

Zetaris

- Three SELECTs build the memory graph: 403 records, the same row counts as the source files (225 chat, 127 tickets, 51 docs)
- The Verifier re-reads each cited record by id; median 0.20 s (chat) and 0.38 s (docs) in the 20 live runs of 24 Oct
- Absence: one UNION ALL query across the three systems
- Stop-gate check on 22 Oct, a script run before the agents were written: cross-system JOIN 45 of 45 rows; 8 of 8 quotes with one word changed were rejected

Meterless Nightshift

- The main model provider for both agents; each agent has its own app id, allowed one model
- 20 live runs: 95 answered model calls, 95 signed receipts, $0 each, 0 routing mismatches (1 more call came back empty and the code took that step)
- Every response is checked: routed model, lane, cost 0; a mismatch stops the run, it is not hidden
- Receipts signed (Ed25519) by the Nightshift daemon, one file per run, e.g. `logs/receipts-20261024T133813Z-5253.jsonl`

Meterless H-MEM + NVIDIA

- H-MEM keeps answers that passed the gate and puts matching ones in front of the model before it answers
- Each Verifier verdict on a recalled memory is feedback: "helpful" adds 0.05 to its confidence. A new memory starts at 0.70; in the cross-session proof one verdict took it to 0.75
- Nemotron 3.5 Lightning: Investigator, 59 of 59 answered calls. Nemotron 3 Nano: Verifier, 36 of 36 (routed model, from the receipts)

WHERE THE TIME GOES IN A LIVE RUN (MEDIAN PER CALL, 20 RUNS)

Zetaris re-read by id	Verifier judgement (Nano)	Investigator tool step (Lightning)	Investigator cites (Lightning)	Investigator revises (Lightning)	Switching the loaded model
0.20 s chat · 0.38 s docs	**2.5 s**	**4.3 s**	**11.9 s**	**44.1 s**	**5.5 s**

Sources: docs/OPEN-AGENT-NEW-WORK.md (Measured), zetaris_spike/README.md, docs/METERLESS.md

Cursor: not used; the organizer says it is optional

The Verifier trusts the source, not the memory and not a model

Before any answer counts, the Verifier goes back through Zetaris to the system that owns each cited record and reads the line again. That check is plain code, and no model can overrule it.

On 24 Oct, in a development run before we moved to Nightshift (Nemotron 3 Nano on a local Ollama), the Investigator's model quoted a chat sentence as if it were ADR-0009, and the Verifier's model said "supports". The quote check rejected it anyway, and the revision quoted the ADR's own line.

Only answers whose every quote is still in its source reach H-MEM, so the agents' memory does not learn from a misquoted answer.

Tampered source: 5 live runs, 24 Oct (Zetaris, Nightshift, H-MEM)

msg-0148 edited in Postgres after the memory read it

- Round 1: `quote_not_in_record`; the model's note: "Sana Petrov remains the owner"
- Round 2: verified on ADR-0009; msg-0148 flagged on its own conflict line
- Stored; the edit restored at the end

5 of 5 runs: round 1 rejected msg-0148, round 2 verified, restored. Median 164.5 s per question: the models think on every call and only one of the two was loaded at a time on our shared Mac. The film shows a later run of the same example (367 s), with the waits between outputs shortened.

A chat assistant over uploaded files (e.g. ChatGPT with file uploads)

Answers from the files it was given. It has no step that reads each quoted line again in the live system before the answer counts.

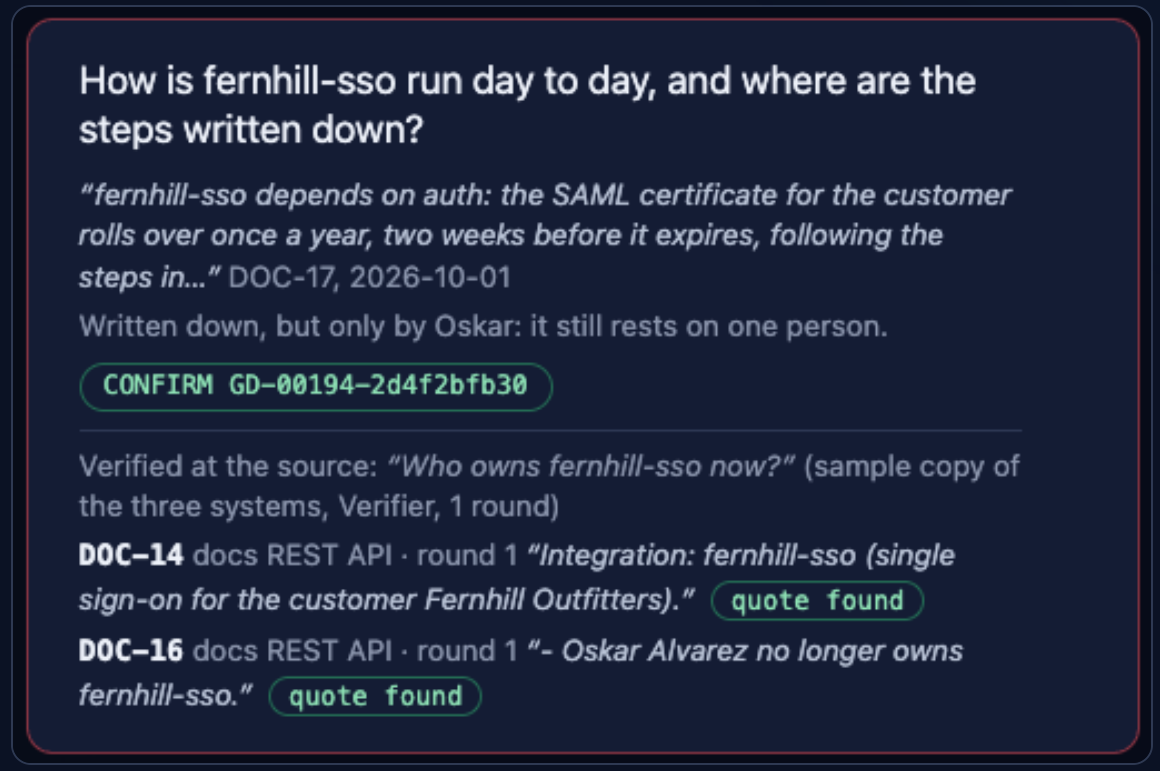
Wiki search (e.g. Confluence search)

Finds pages. The lead reads and reconciles them by hand; nothing reads a quoted line again at its source.

What Leaves With Them

Every cited claim is re-read at its source before it counts, what could not be confirmed is listed, and only checked answers are stored.

One question, on screen and in the trace



The answer card, cropped from a real screenshot of the panel (sample mode). The card asks "How is fernhill-sso run day to day, and where are the steps written down?". Under the divider, the panel asked the agents "Who owns fernhill-sso now?" and shows the two records the Verifier re-read in the sample copy of their source system; this part was built in the hackathon window.

```
docs/deck-runs-2026-10-25/
trace-20261025T135528Z-6fd1.jsonl
same question, sample mode (a local copy of the three
systems), on the submitted code; abridged, from the trace

 1 Run          start        "Who owns fernhill-sso now?"
 5 Ingest       graph_built  403 records, 3 systems  1.23 s
 7 Investigator memory_recall  H-MEM: nothing matched
 8 Investigator tool_call    current_owner fernhill-sso
                             → DOC-14, DOC-16
10 Investigator draft        round 1, 2 citations → Verifier
12 Verifier     reread       DOC-14  docs (sample copy)
13 Verifier     check        DOC-14  found=True  accepted
14 Verifier     reread       DOC-16  docs (sample copy)
15 Verifier     check        DOC-16  found=True  accepted
16 Verifier     verdict      round 1 accepted → write gate
18 Investigator write_gate   accepted  FND-00001, H-MEM mem-1
19 Run          end          1.34 s  verified

Answer: "fernhill-sso has no current owner. Last owner:
Oskar Alvarez until 2026-10-09 (closed by GD-00225-...,
citing DOC-16). Nobody has been recorded as taking it over."
No model in sample mode, so the model's judgement reads
"skipped"; live runs add each model call with its signed
Nightshift receipt id (slide 5, logs/).
```

Next: real systems, same rule

In the product

- Refresh the memory graph when the Verifier finds a record changed at its source (today the catch is reported and the graph keeps the old copy).
- A Zetaris semantic layer with a read-only agent user, so the agents see only the views they need.
- Absence checks that ignore letter case, and recall in H-MEM by meaning, not only by shared words.
- Faster live runs: both Nemotron models loaded at once and thinking switched off per call (today a median of 18.4 s to 164.5 s per question).

With real teams

- More sources behind Zetaris: meeting notes, end dates from the HR system, code ownership files.
- A pilot on a real team's records, measured against the same benchmark questions.
- Handover cards that the successor confirms, written back through the same gate.

Limits today

Source edits were tested on chat records only. H-MEM recall matches shared words, and the absence check is case-sensitive. Answer quality is bounded by the model: the guarantee is about what gets stored, not that every question gets answered.

Check the quote before you trust the answer.